Power BI Project Report

Chocolate Sales Dashboard • Business Intelligence Portfolio

Tool: Microsoft Power BI Desktop • File: choclate_sales_dashboard.pbix Prepared by Abhishek Chauhan

Executive Summary

This report documents the Chocolate Sales Dashboard built in Microsoft Power BI Desktop. The dashboard provides a single-page interactive view of chocolate sales performance across products, customer demographics, and store locations — enabling sales managers to monitor KPIs, track monthly profit trends, identify top-performing products, and filter results by country, city, and store type.

Tool: Microsoft Power BI Desktop **Visuals:** 7 (4 KPI Cards, 1 Line, 1 Column, 1 Treemap, 3 Slicers) **Tables:** 5 (measures, calendar, customers, products, stores)

Dashboard Overview

The Chocolate Sales Dashboard is a single-page Power BI report that gives sales management a complete commercial picture at a glance. It combines headline KPIs, time-series profit trends, customer age segmentation, and product revenue breakdown — all filterable by geography and store type through three interactive slicers.

#	Visual Type	Fields Used	Business Question Answered	Category
1	KPI Card (x4)	Total Profit, Total Sale, Total Customers, Total Products	What are the headline KPIs?	KPI
2	Line Chart	MoM Profit %, Date Hierarchy (Year/Month)	How is profit trending month over month?	Trend
3	Clustered Column	Age Group (customers), Total Sale	Which customer age group drives the most sales?	Audience
4	Treemap	Product Name, Category, Total Sale	Which products & categories generate revenue?	Product
5	Slicer — Country	stores.country	Filter all visuals by country	Filter
6	Slicer — City	stores.city	Filter all visuals by city	Filter
7	Slicer — Store Type	stores.store_type	Filter all visuals by store type	Filter

Section 1 — KPI Cards (Headline Metrics)

Four KPI cards sit at the top of the dashboard, giving stakeholders an instant read on the business's four most critical numbers. These are the first thing any viewer sees and set the context for all other visuals.

Visual 1 • KPI Card Visual (x4) [Card]

Purpose	Display the four top-level business KPIs as large, prominent numbers: Total Profit, Total Sale, Total Customers, and Total Products — giving executives an immediate health-check of the business.
Business Value	A single glance tells the leadership team whether the business is on track. These four numbers answer: Are we profitable? How much are we selling? How many customers do we have? How broad is our range?
Key Insight	**All four cards pull from the 'measures table' — a dedicated DAX measures table keeping calculations separate from raw data. This is Power BI best practice: one table, all logic.**

Fields & Configuration

```
Visual Type : cardVisual (x4)
Data Table  : measures table (DAX)

Card 1 — Total Profit
  Measure : measures table[total profit]
  Format  : Currency / Decimal

Card 2 — Total Sale
  Measure : measures table[Total sale]
  Format  : Currency / Decimal

Card 3 — Total Customers
  Measure : measures table[Total Customers]
  Format  : Whole Number

Card 4 — Total Products
  Measure : measures table[Total Products]
  Format  : Whole Number
```

Section 2 — Profit Trend Analysis

The line chart tracks month-over-month profit percentage change over time, helping management spot growth momentum, seasonal dips, and inflection points in business performance.

Visual 2 • Line Chart [Line]

Purpose	Visualise the Month-over-Month (MoM) profit percentage change over the full date range, drilled by Year and Month from the calendar date hierarchy.
Business Value	Reveals whether profit is accelerating or decelerating month by month. Identifies seasonal patterns in chocolate sales (e.g. festive peaks, summer troughs) and validates whether strategic initiatives are improving profitability.
Key Insight	**Uses a date hierarchy (Year > Month) from the calendar table enabling drill-down from annual to monthly view. The MoM % measure compares each month to the prior month — a time-intelligence calculation that requires the calendar table relationship.**

Fields & Configuration

```
Visual Type : lineChart
X-Axis      : calendar[date] → Date Hierarchy
              └── Year  (drill level 1)
              └── Month (drill level 2)

Y-Axis      : measures table[mom_profit %]
```

```
                (Month-over-Month profit % change)

Related Table : calendar
Relationship  : calendar[date] → fact table date column
```

Section 3 — Customer & Product Analysis

These two visuals break down sales by who is buying (age group) and what they are buying (products and categories) — the core of any retail analytics dashboard.

Visual 3 • Clustered Column Chart [Bar]

Purpose	Show total sales broken down by customer age group — revealing which demographic segments are the highest-value buyers of chocolate products.
Business Value	Tells marketing which age groups to target with campaigns and promotions. If the 35-54 age band dominates sales, campaigns should be weighted towards that demographic for maximum ROI.
Key Insight	**Uses customers[age group] as the X-axis category and measures table[Total sale] as the Y-axis value — a simple but powerful cross-table relationship that requires a relationship between the customers table and the fact/sales data.**

Fields & Configuration

```
Visual Type : clusteredColumnChart
X-Axis (Category) : customers[age group]
Y-Axis (Value)    : measures table[Total sale]

Data Tables Used  : customers, measures table
Expected Age Groups (typical segmentation):
  < 18   | 18-24 | 25-34 | 35-54 | 55+
```

Visual 4 • Treemap [Tree]

Purpose	Display revenue contribution by product and product category using proportional rectangles — making it immediately clear which products and categories dominate sales by visual area.
Business Value	Product managers can see at a glance which SKUs are carrying the portfolio and which are dead weight. Category managers use this to make ranging and delisting decisions. The bigger the rectangle, the bigger the revenue contribution.
Key Insight	**Treemap uses products[category] as the Group and products[product_name] as the Detail, with measures table[Total sale] driving the size of each rectangle. Hierarchical view: outer rectangles = categories, inner = individual products.**

Fields & Configuration

```
Visual Type  : treemap
Group        : products[category]
               (outer rectangle = product category)
Detail       : products[product_name]
               (inner rectangle = individual product)
Values       : measures table[Total sale]
               (rectangle size = revenue)

Data Tables  : products, measures table
```

Section 4 — Interactive Slicers (Filters)

Three slicers allow users to filter the entire dashboard by store geography and type — turning a static report into a fully interactive analytical tool. Every visual on the page responds to these filters simultaneously.

Visual 5 • Slicer — Country [Filter]

Purpose	Allow users to filter the entire dashboard to one or more countries — isolating performance metrics, trends, and product data for a specific geographic market.
Business Value	A regional director can select their own country and instantly see KPIs, profit trends, and product sales that are relevant only to their market — without needing a separate report.
Key Insight	**Connected to stores[country]. Because the stores table relates to the fact data, selecting a country cascades through all relationships — every visual on the page filters automatically.**

Fields & Configuration

```
Visual Type : slicer
Field       : stores[country]
Slicer Mode : List or Dropdown
Cross-filter: Affects ALL visuals on the page
              → KPI Cards, Line Chart, Column Chart, Treemap
```

Visual 6 • Slicer — City [Filter]

Purpose	Enable city-level filtering of the dashboard — allowing users to drill down from country to specific cities for granular local market analysis.
Business Value	Store managers and city-level sales teams can view performance for their specific city. Useful for comparing high-performing cities within the same country — e.g. Mumbai vs Delhi.
Key Insight	**Connected to stores[city]. Works in combination with the Country slicer for cascaded filtering: select a country first, then narrow to a specific city within it for precise local intelligence.**

Fields & Configuration

```
Visual Type : slicer
Field       : stores[city]
Slicer Mode : List or Search (recommended for many cities)
Cross-filter: Affects ALL visuals on the page
Best Practice: Use with Country slicer for cascaded filtering
```

Visual 7 • Slicer — Store Type [Filter]

Purpose	Filter the dashboard by the type of retail store — enabling comparison of chocolate sales performance across different distribution channels (e.g. supermarket vs specialist vs online).

Business Value	Reveals which store formats drive the most chocolate sales. If premium chocolate sells better in specialist stores than supermarkets, it informs ranging strategy, pricing, and new store partnership decisions.
Key Insight	**Connected to stores[store_type]. Slicing by store type answers the channel strategy question: where should we focus distribution investment to maximise chocolate revenue?**

Fields & Configuration

```
Visual Type : slicer
Field       : stores[store_type]
Slicer Mode : List or Dropdown
Cross-filter: Affects ALL visuals on the page
Example Types (typical): Supermarket | Specialist
                         Convenience | Online | Department
```

Data Model

The Power BI data model contains 5 tables. The 'measures table' acts as a centralised DAX calculation layer, while the remaining 4 tables are dimension tables that provide categories and hierarchies for filtering and grouping.

Table	Key Columns	Role in Model	Visual Used In
measures table	total profit, Total sale, Total Customers, Total Products, mom_profit %	DAX measures hub — all KPIs and calculations live here	**All visuals**
calendar	date (hierarchy: Year, Month)	Date dimension enabling time-intelligence calculations like MoM %	**Line Chart**
customers	age group	Customer dimension — demographic segmentation	**Column Chart**
products	product_name, category	Product dimension — name and category hierarchy	**Treemap**
stores	country, city, store_type	Store dimension — geographic and store-type filtering	**All 3 Slicers**

DAX Measures Reference

All 5 DAX measures are stored in the dedicated 'measures table'. This separation of logic from raw data is Power BI best practice — it keeps the data model clean and all calculations in one auditable location.

Measure Name	What It Calculates	Used In
Total Sale	Sum of all sales revenue across all products and stores	KPI Card, Column Chart, Treemap
Total Profit	Sum of profit earned after costs — the bottom-line performance metric	KPI Card
Total Customers	Count of distinct customers who have made purchases	KPI Card
Total Products	Count of distinct products sold or listed in the catalogue	KPI Card

Measure Name	What It Calculates	Used In
`mom_profit %`	Month-over-Month profit percentage change — compares this month to last month	Line Chart

Power BI Features & Techniques Used

A summary of all Power BI concepts and features applied in this dashboard project.

Feature / Technique	Description
DAX Measures Table	Centralised table storing all KPI calculations separately from raw data — best practice for maintainability
Time Intelligence	MoM profit % uses calendar table + date hierarchy for month-over-month comparison calculations
Date Hierarchy	calendar[date] broken into Year > Month levels enabling drill-down in the Line Chart
Cross-filtering	All 3 slicers cross-filter every visual on the page simultaneously via data model relationships
Treemap Visual	Hierarchical area chart showing category (group) and product (detail) revenue in a single visual
KPI Card Visual	Large-format metric display for headline numbers — the fastest way to communicate business health
Clustered Column	Side-by-side bars for categorical comparison — used here for age group vs total sales
Slicer Visual (x3)	Interactive filter controls for country, city, and store type — enables self-service analysis for end users
Star Schema Design	measures table at centre + 4 dimension tables (calendar, customers, products, stores) radiating outward

Report Complete 7 visuals • 5 tables • 5 DAX measures • 1 page